

14-amaliy mashg'ulot. Sahna mazmuni, scene graph va izoh

2026–2027 o'quv yili

Mavzu. Multi-label tasniflash (sigmoid va softmax, BCE, focal loss, ostona); metrikalar (precision, recall, F1, macro/micro, exact match, Hamming loss); CAM va Grad-CAM, CAM dan bounding box; scene graph va shablonli image captioning; BLEU, so'z F1; CLIP (InfoNCE, zero-shot), beam search.

Maqsadlar. Talaba: (1) multi-label va scene graph metrikalarini, CAM, BLEU va CLIP ehtimollarini qo'lda hisoblaydi; (2) BCE gradienti, softmax cheklovi, CAM-logit tengligini isbotlaydi; (3)

PyTorch/scikit-learn/OpenCV da sun'iy sahnalarda to'liq zanjirni yozadi: tasniflash → CAM → scene graph → izoh → baholash.

Baholash (100 ball). A (1–11): $11 \times 2 = 22$; B (12–16): $5 \times 3 = 15$; C (17–30): $14 \times 4,5 = 63$. Kodda: to'g'ri ishlashi — 60%, chiqish formati va sonlar — 25%, tahlil — 15%.

Muhit. Python 3.11; requirements.txt: numpy==2.4.4, scikit-learn==1.8.0, opencv-python-headless==4.13.0.92, torch==2.14.0. Ma'lumotlar sun'iy (make_multilabel_classification va quyidagi sahna generatori), seed=42, faqat CPU, torch.set_num_threads(1). python -W error::DeprecationWarning da ogohlantirish bo'lmasin.

Hujjatlar. https:

https://pytorch.org/docs/stable/generated/torch.nn.functional.binary_cross_entropy_with_logits.html,
https://pytorch.org/docs/stable/generated/torch.Tensor.retain_grad.html, https://scikit-learn.org/stable/modules/generated/sklearn.datasets.make_multilabel_classification.html,
https://scikit-learn.org/stable/modules/generated/sklearn.metrics.f1_score.html,
https://docs.opencv.org/4.x/d3/dc0/group__imgproc__shape.html.

1. A bo'lim. Qo'lda hisoblash

- 3 sinf: $TP = (30, 18, 9)$, $FP = (5, 2, 6)$, $FN = (4, 6, 3)$. Har sinf uchun precision, recall, F1; macro va micro F1. Nega micro macro dan katta?
- 6 namuna, 4 yorliq: Y qatorlari 1010, 0110, 1100, 0001, 1011, 0100; bashorat $\hat{Y}$: 1010, 0100, 1110, 0001, 1001, 0000. Exact match, Hamming loss, micro F1 va namunalar bo'yicha o'rtacha F1 (bo'sh-bo'sh juftlik yo'q).
- Logitlar $z = (1,5; 0,5; -1; 1,2)$. Sigmoid va softmax ehtimollari; $\tau = 0,5$ da har birida nechta yorliq? Softmax da $\tau = 0,3$ da ko'pi bilan nechta yorliq bo'lishi mumkin va aslida nechta?
- $z = (2; -1; 0,5)$, $y = (1, 0, 1)$: BCE (yig'indi), gradient $\partial \mathcal{L} / \partial z$; pos_weight=3 bo'lganda loss va gradient.
- Focal loss ($\gamma = 2$): $p_t = 0,8$ va $p_t = 0,3$ da CE va FL; oson/qiyin nisbati qanday o'zgardi? Uchrash chastotasi 8% bo'lgan sinf uchun pos_weight?
- CAM: $f_1 = \begin{pmatrix} 0 & 2 & 1 \\ 0 & 3 & 0 \\ 0 & 0 & 0 \end{pmatrix}$, $f_2 = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 0 \\ 2 & 0 & 1 \end{pmatrix}$, $w_c = (1; -0,5)$, $b_c = -0,3$. CAM_c, cho'qqi, GAP orqali z_c , CAM o'rtachasi $+b_c$, $\sigma(z_c)$.
- Kirish 96×96 , uchta 2×2 pooling: belgi xaritasi o'lchami va stride S ; cho'qqi katagi $(i, j) = (4, 9)$ ning piksel markazi (x, y) ; maksimal kvantlash xatosi. ViT (224 , $P = 16$) da tokenlar soni va fazoviy aniqlik?
- Markazlar: kvadrat (20, 40), doira (22, 12), uchburchak (50, 42), yulduz (47, 15) (tasvirda y pastga). Qoida: b ning a ga nisbatan o'rni, $|\Delta x|, |\Delta y| < 3$ — “ustma-ust”, $|\Delta x| > |\Delta y|$ — “o'ng/chap tomonida”, aks holda “pastida/ustida”. Ro'yxat tartibida barcha $(b, \text{predikat}, a)$ uchliklarni yozing. Qaysi uchlik beqaror? 5 predikatda nomzod uchliklar soni?
- 8-masaladagi 6 uchlik — haqiqiy graf. Model ishonch bo'yicha tartiblangan uchliklar: (doira, ustida, kvadrat), (yulduz, o'ng, doira), (uchburchak, ustida, doira), (uchburchak, o'ng, kvadrat), (yulduz, ustida, kvadrat), (yulduz, ustida, uchburchak). Recall@3, Recall@6, precision va F1 (6 ta bo'yicha).
- Ma'lumotnoma: “sahnada doira kvadratning ustida joylashgan”. Bashoratlar: (a) “sahnada doira kvadratning pastida joylashgan”, (b) “doira kvadratning ustida”. Har biri uchun so'z F1, p_1 , p_2 , BP va BLEU-2. Xulosa?
- CLIP: kosinuslar (0,28; 0,25; 0,21); $1/\tau = 100$ va $1/\tau = 10$ da zero-shot ehtimollar. $N = 2$ batch, masshtablangan o'xshashliklar $S = (\frac{5}{2} \frac{1}{4})$: simmetrik InfoNCE; tasodifiy holatdagi $\ln N$ bilan solishtiring.

2. B bo'lim. Isbot va tushuncha

- BCE uchun $\partial \mathcal{L} / \partial z_c = \sigma(z_c) - y_c$ ni isbotlang. pos_weight= w da gradient qanday? Nega bu sinflarni mustaqil ikkilik masalalarga ajratadi?
- Softmax da τ ostonadan o'tgan sinflar soni $k < 1/\tau$ ekanini isbotlang. Nega “ τ ni kamaytirish” ko'p yorliqli vazifaga yechim emas?

14. GAP + chiziqli tarmoqda $z_c = \frac{1}{HW} \sum_{x,y} \text{CAM}_c(x,y) + b_c$ va Grad-CAM = $\text{ReLU}(\text{CAM}_c)/(HW)$ ekanini isbotlang.
15. $P = 1$ da $F1 = 2R/(1 + R)$. Har yorliq mustaqil a aniqlikda bo'lsa, exact match = a^C ; $C = 3$, $a = 0,99$ da? Macro va micro F1 qachon keskin farq qiladi?
16. Scene graph baholashda nega precision emas, Recall@K ishlatiladi? Nega BLEU “chap” $\leftrightarrow$ “o'ng” xatosini kuchsiz sezadi va SPICE (graf uchliklari bo'yicha F1) nimasi bilan yaxshi? Tasodifiy CLIP da $\mathcal{L} \approx \ln N$ ekanini tushuntiring.

3. C bo'lim. Dasturlash (PyTorch / scikit-learn / OpenCV)

Har topshiriq — alohida funksiya; main() ularni tartib bilan chaqiradi va [raqam] ... formatida chop etadi. 22–27-topshiriqlar uchun umumiy sahna generatori va tarmoq:

```
import numpy as np, cv2, torch, torch.nn as nn, torch.nn.functional as F
torch.set_num_threads(1); SEED = 42; NAMES = ["to'rtburchak", "doira", "uchburchak"]
def sahna(rng, S=32):
    # 3 shakl (p=0.6), har biri alohida kvadrantda, siljish 1..5 px
    img = np.zeros((S, S), np.float32); y = (rng.random(3) < 0.6).astype(np.float32)
    quad = rng.permutation(4)[:3]; objs = []
    for c in range(3):
        ox, oy = rng.integers(1, 6, 2)
        if not y[c]: continue
        x0, y0 = int((quad[c] % 2) * 16 + ox), int((quad[c] // 2) * 16 + oy)
        if c == 0: cv2.rectangle(img, (x0, y0), (x0 + 9, y0 + 9), 1, -1)
        elif c == 1: cv2.circle(img, (x0 + 5, y0 + 5), 5, 1, -1)
        else: cv2.fillPoly(img, [np.array([[x0, y0+10], [x0+10, y0+10], [x0+5, y0]], np.int32)], 1)
        ys, xs = np.nonzero(img[y0:y0 + 11, x0:x0 + 11]) # og'irlik markazi va maydon
        objs.append(dict(c=c, nom=NAMES[c], cx=float(xs.mean() + x0), cy=float(ys.mean() + y0),
                        s=int(len(xs)), box=(x0, y0, x0 + 10, y0 + 10)))
    return img + rng.normal(0, 0.1, img.shape).astype(np.float32), y, objs
# rng = np.random.default_rng(42); avval 800 ta o'quv, keyin 300 ta test sahnasi
class Net(nn.Module):
    # conv -> GAP -> linear (CAM mumkin)
    def __init__(self):
        super().__init__()
        self.features = nn.Sequential(nn.Conv2d(1, 16, 3, padding=1), nn.ReLU(), nn.MaxPool2d(2),
                                      nn.Conv2d(16, 32, 3, padding=1), nn.ReLU(), nn.MaxPool2d(2))
        self.fc = nn.Linear(32, 3)
    def forward(self, x): return self.fc(self.features(x).mean((2, 3)))
```

17. multilabel_metrics(Y, P) -> dict (P, R, F1 har sinf; macro, micro, exact, hamming; nolga bo'lishdan himoya). Ma'lumot: rng=default_rng(42), Y=(rng.random((50,4))<[0.6,0.4,0.2,0.1]), P=np.where(rng.random(Y.shape)<0.15, 1-Y, Y). sklearn.metrics bilan tekshiring. Chiqish: [17] F1=[...] macro=... micro=... exact=... hamming=... va sklearn: ...
18. Use-case: multi-label tasniflash. make_multilabel_classification(n_samples=2000, n_features=20, n_classes=5, n_labels=2, random_state=42); 1000/500/500 (o'quv/val/test). (a) har sinfga LogisticRegression(max_iter=2000) (sigmoid, OvR); (b) “softmax”: har musbat yorliq alohida namuna, bitta multinomial LogisticRegression. $\tau = 0,5$ da test metrikalari va yorliq/namuna. Chiqish: [18] name macro F1=... micro F1=... exact=... hamming=... labels/sample=...
19. 18(a) modeli uchun har sinf ostonasini validatsiyada np.arange(0.05,0.96,0.05) to'ridan F1 maksimumi bo'yicha tanlang; test macro/micro F1 ni $\tau = 0,5$ bilan solishtiring. Chiqish: [19] tau_c=[...]; test macro F1: tau=0.5 -> ..., tuned -> ...; micro ... -> ...
20. F.binary_cross_entropy_with_logits (sum) va autograd bilan 4-masalani tekshiring: $\max |\nabla - (\sigma - y)|$; pos_weight=3. Chiqish: [20] BCE sum=... grad=[...] max|grad-(sigma-y)|=...; pos_weight=3: loss=... grad=[...].
21. focal_loss(z, y, gamma) (o'rtacha). $\gamma = 0$ da BCE ga tengligini tasodifiy $z=3*\text{torch.randn}(1000)$, $y=\text{torch.rand}(1000)<0.5$ (manual_seed(42)) da tekshiring; $y = 1$, $p_t = 0,9$ va $0,4$ da CE, FL va oson/qiyin nisbatlari. Chiqish: [21] |FL(gamma=0)-BCE|=...; p_t=0.9: CE=... FL=...; p_t=0.4: ...; ratio easy/hard CE=... FL=...
22. Use-case: sahna tasnifi. Net ni 25 epoxa, batch 32, Adam(lr=3e-3), manual_seed(42) bilan ikki xil o'qiting: sigmoid + BCE va softmax + yumshoq nishon $-\sum_c \frac{y_c}{\max(1, |y|)} \log \text{softmax}(z)_c$. Test: $\tau = 0,5$ da macro F1, exact, yorliq/sahna. Chiqish: [22] kind macro F1=... exact=... labels/scene=... (true ...).
23. Sigmoid tarmoq uchun CAM = $\sum_k w_{c,k} f_k$ (einsum); $\max |z - (\text{CAM} + b)|$. Har haqiqiy obyekt uchun o'z sinfi CAM cho'qqisi katagi markazi ($Sj + \frac{S-1}{2}, Si + \frac{S-1}{2}$) va og'irlik markazi orasidagi masofa: o'rtacha, mediana, $\leq S$ va $\leq 2S$ ulushi, tasodifiy nuqta (uniform(0,32,2), rng(42)) bilan. Chiqish: [23] max|logit-(mean

- CAM+b)|=...; stride S=...; objects=... mean dist=... px median=...
24. Grad-CAM ni `retain_grad` bilan (1-test tasviri, $\sin c = 1$) hisoblang va $\text{ReLU}(\text{CAM})/(HW)$ bilan solishtir-ing. *Chiqish:* [24] `max|GradCAM - ReLU(CAM)/(HW)|=...; same peak=...`
 25. **CAM $\rightarrow$ bounding box.** CAM ni $[0, 1]$ ga normallang, `cv2.resize(..., (32,32), INTER_LINEAR)`, $\geq \theta$ maska, `cv2.connectedComponentsWithStats` (8-bogʻlanish) boʻyicha eng katta komponent qutisi; haqiqiy box bilan IoU (piksel inklyuziv). $\theta \in \{0,2; 0,5; 0,7\}$ da oʻrtacha IoU va CorLoc. *Chiqish:* [25] `theta=...: mean IoU=... CorLoc(IoU>=0.5)=...%` (3 qator).
 26. `munosabat(a, b, tol=3)` (8-masala qoidasi), `scene_graph(objs)` (sinf tartibida juftliklar) va `caption(objs)` (“Sahnada 3 ta obyekt bor: ...; doira toʻrtburchakning oʻng tomonida; ...; eng kattasi - ...”). Uch obyektli birinchi test sahnasi va test boʻyicha predikatlarni sanogʻi. *Chiqish:* [26] `test scene #k: objects=..., uchliklar, izoh va predicate counts.`
 27. **Use-case: sahnadan izohga.** Sigmoid tarmoq ($p_c > 0,5$) + CAM choʻqqisi $\rightarrow$ bashorat obyektlari $\rightarrow$ scene graph. Test boʻyicha uchliklar TP/FP/FN, P, R, F1; “eng kattasi” qismisiz izohlar uchun oʻrtacha soʻz F1 va toʻliq mos izohlar ulushi. *Chiqish:* [27] `triplets: TP=... FP=... FN=... P=... R=... F1=...; caption word F1=... exact caption=...%`
 28. `bleu(ref, hyp, N)` (qirgʻilgan n -gram precision, BP, geometrik oʻrtacha) va `word_f1`. Maʼlumotnoma “sahnada 2 ta obyekt bor doira uchburchakning chap tomonida”; bashoratlar: “...doira uchburchakning ustida”, “...doira uchburchakning oʻng tomonida”, “sahnada 2 ta obyekt bor”. *Chiqish:* [28] `|hyp|=.. word F1=... p1=... p2=... BLEU-2=... BLEU-4=... (3 qator).`
 29. **CLIP.** `inforce(U, V, scale)` (simmetrik). `rng=default_rng(42)`, $N = 256$, $d = 512$: tasodifiy juftlar (scale 1) va $V=U+N(0,1)$ (scale 10). Zero-shot: 10 sinf, $T=\text{base}+0.4*N(0,1)/\text{sqrt}(d)$ ($\text{base}=N(0,1)/\text{sqrt}(d)$), normallang; 1000 tasvir $I=T[\text{lab}]+N(0,0.15)$ (normallangan). Anqlik, oʻrtacha kosinuslar, $\text{scale} \in \{1, 10, 100\}$ da oʻrtacha max ehtimol. *Chiqish:* [29] `InfoNCE random=... (ln N=...) aligned=..., zero-shot acc=...` va 3 qator.
 30. **Beam search.** Lugʻat `<s>`, doira, kvadrat, ustida, pastida, `</s>`; oʻtish ehtimollari (qatorlar — joriy soʻz): `<s>`: doira 0,55, kvadrat 0,45; doira: ustida 0,5, pastida 0,3, `</s>` 0,2; kvadrat: ustida 0,05, pastida 0,05, `</s>` 0,9; ustida/pastida: doira 0,2, kvadrat 0,2, `</s>` 0,6. Greedy va beam ($B = 2, 3$), $T \leq 4$. *Chiqish:* [30] `name: jumla P=... mean log P/token=...`